Do larger molecular models improve antimalarial activity prediction? A scaffold-controlled comparison with ECFP4

Myke Vital Sao Temgoua*[1], Jean-Pierre Tchapet Njafa[1], Serge Guy Nana Engo[1], Penabei Samafou[2], and Wilfred Fon Mbacham[3]

[1]Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon
[2]Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada
[3]Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

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Abstract

On a curated antimalarial natural-product panel, do graph neural networks and pretrained sequence transformers improve activity ranking beyond a compact ECFP4–random-forest baseline when chemical scaffolds are held out? We address this question on a curated panel of 19 836 African antimalarial natural products assembled from the eOS80CH screening collection, using random and Bemis–Murcko scaffold five-fold cross-validation, each repeated over five seeds (25 fold–seed replicates per split). We compare a random-forest baseline on ECFP4 fingerprints against a graph isomorphism network (GIN), two topological fusion variants (GIN–TFP and GIN–TNE built on persistent-homology and tensor-network descriptors), and a fine-tuned ChemBERTa sequence transformer. Under the scaffold split — the practically relevant out-of-distribution setting — ECFP4–RF achieves the highest ROC AUC (0.830 0), followed by GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7), and ChemBERTa (0.786 7). No GNN or transformer arm outperforms the fingerprint baseline; under the scaffold split *every* arm is significantly worse (GIN–TFP $\Delta = -0.016$, raw $p = 0.025$, BH-adjusted $p = 0.0330$; GIN–TNE $\Delta = -0.021$, raw $p = 0.038$, BH-adjusted $p = 0.0378$; GIN $\Delta = -0.025$, raw $p = 0.018$, BH-adjusted $p = 0.0330$; ChemBERTa $\Delta = -0.043$, raw $p < 0.0001$, BH-adjusted $p = 0.0002$). Under the random split the same verdict holds: every arm is significantly below ECFP4–RF (GIN–TFP 0.908 4, GIN–TNE 0.891 8, GIN 0.909 8, ChemBERTa 0.912 1 versus 0.943 3; all raw $p < 0.0001$). The transformer was initialized independently from the pretrained state at the beginning of every fold, preventing transfer of fine-tuned parameters between training folds; all reported transformer results use this fold-independent protocol. Finally, an analysis of the fusion head shows that persistent-image dimensions of the topological fingerprints have the largest mean absolute projection weights, providing a hypothesis-generating link to molecular geometry. The results show that representation–task alignment, rather than model scale alone, governs performance on this panel. Code, frozen splits, and trained checkpoints are released.

Contribution: We provide a controlled benchmark showing that ECFP4–RF remains the strongest tested predictor on an antimalarial natural-product panel under scaffold separation, while topological fusion supplies descriptive geometric signal without improving ranking. The

*Corresponding author: myke-vital.sao@facsciences-uy1.cm

study also specifies fold-independent initialization for the pretrained transformer, a necessary condition for valid cross-validation.

Keywords: graph neural networks; molecular fingerprints; antimalarial natural products; scaffold split; benchmark; interpretability; persistent homology

Highlights:

- Under the scaffold split, ECFP4–RF (AUC 0.830 0) outperforms every GNN/transformer arm; all arms are significantly worse (BH-adjusted $p \leq 0.037\,8$)
- Fold-independent transformer initialization changes the apparent ranking of sequence models and is required for valid cross-validation
- Persistent-image dimensions have the largest descriptive projection-weight salience in the topological fusion head
- Reproducible frozen splits (random + scaffold), 25 fold–seed replicates per split, and released checkpoints

1 Introduction

Malaria remains a leading infectious disease burden in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2 023 [1]. Control is increasingly threatened by the emergence of artemisinin-resistant *Plasmodium falciparum* strains in Southeast Asia and Africa [2, 3], prompting an urgent, WHO-endorsed need for new scaffolds that remain active against drug-resistant parasites. African natural products (NPs) constitute a rich source of chemically diverse scaffolds [4, 5]. The panel studied here descends from a screening campaign seeded on 396 African NP and 454 synthetic-drug scaffolds [6]; separate molecular-dynamics work in the same research programme examined selected leads against resistance-associated targets and mutations [7]. Computational screening of NP libraries promises to accelerate hit identification in this resistance-driven search, yet the cheminformatics representation strongly conditions the outcome. In recent years, a "scale-centric" narrative has emerged in molecular machine learning: pretrained molecular foundation models and sequence transformers are expected to supersede compact, task-trained cheminformatics models [8, 9]. This narrative extends to quantum-inspired representations, where quantum computing is increasingly discussed as a paradigm for drug discovery and bioinformatics [10, 11].

This narrative is increasingly challenged by rigorous, controlled benchmarks. In a 156-comparison benchmark across 26 endpoints, Guo and Ding found that classical machine learning on fingerprints wins 47.4 % of task–metric entries, ahead of pretrained sequence models (28.8 %) and GNNs (21.8 %), with the classical advantage largest under random splits and shrinking, but not vanishing, under scaffold-separated protocols [12]. An independent comparison of 25 pretrained embedding models across 25 datasets found that *almost no neural embedding statistically improves over the ECFP fingerprint baseline* [13]. And for natural products specifically, a 20-fingerprint benchmark over 100 000+ NPs showed that no single encoding dominates, and that ECFP is frequently matched or exceeded by alternative fingerprints on NP bioactivity tasks [14].

Three questions determine whether this conclusion is informative rather than merely negative. *First*, do the findings extend from drug-like benchmark libraries to a dedicated antimalarial natural-product panel with high structural complexity? *Second*, do persistent-homology and tensor-network descriptors add value inside a modern GNN fusion architecture under scaffold separation? *Third*, does fold-independent initialization alter the apparent performance of a pretrained transformer? These questions motivate the controlled comparisons and the validity protocol used below.

1.1 Related work

The tension between classical cheminformatics and neural molecular learning has been documented across multiple benchmarks. For molecular property prediction, random forests on ECFP fingerprints remain competitive with or superior to GNNs on small-to-medium datasets ($< 10\,000$ molecules) [12, 13]. GNN architectures (GCN, GAT, GIN) have shown gains on large, curated datasets such as MoleculeNet [15], but these gains often diminish under scaffold splits that better reflect real-world drug discovery [12]. Pretrained sequence transformers (ChemBERTa, MolBERT) leverage large unlabeled corpora but have shown inconsistent gains on downstream tasks, particularly when the target dataset is structurally dissimilar to the pretraining data [13, 16].

Topological data analysis (TDA) has been applied to molecular representation through persistent homology, producing descriptors that capture ring systems, molecular cavities, and higher-order connectivity [17]. Recent work has fused TDA features with GNN architectures, reporting modest improvements on specific tasks [18]. Multimodal fusion of graph and sequence representations—for example, concatenating GCN embeddings with ChemBERTa latent vectors—has shown promise on small datasets [19]. Tensor-network embeddings, derived from Tucker decomposition of molecular tensors, provide compressed representations that retain scaffold-relevant information [9]. However, neither TDA nor tensor-network features have been systematically tested inside GNN fusion architectures under rigorous scaffold-split protocols on antimalarial panels.

We therefore benchmark GIN, GIN–TFP, GIN–TNE, and ChemBERTa against ECFP4–RF under both random and scaffold splits. We test three linked questions: whether learned models exceed ECFP4, whether topological fusion narrows the scaffold gap, and whether projection-weight salience identifies a reproducible geometric signal even when ranking does not improve. The comparison is designed to distinguish predictive performance from interpretability and to quantify the effect of fold-independent transformer initialization.

2 Results

2.1 Baseline and evaluation question: a reproducible reference for model comparison

To verify reproducibility, we independently reproduced the ECFP4–RF baseline reported for the same molecular collection [20]. Under the random split we obtain $0.943\,3 \pm 0.000\,3$ across five per-seed means (range 0.942 8–0.943 7), compared with the reference value of $0.947\,5 \pm 0.004\,5$. The difference is compatible with the distinct frozen split and random-seed realization used here, supporting comparability of the panel and fingerprint implementation.

2.2 H1: scaffold-held-out prediction favours ECFP4 over learned representations

table 1 reports the mean test ROC AUC and the SD across five per-seed means for each model and split; each per-seed mean aggregates the five fold-level test AUCs. Under the scaffold split, the fingerprint baseline reaches 0.830 0. Every graph-based and transformer arm is significantly *worse*: the best GNN arm, GIN–TFP, reaches 0.813 8 ($\Delta = -0.016$, raw paired $p = 0.025$; BH-adjusted $p = 0.033\,0$); GIN–TNE reaches 0.809 0 ($\Delta = -0.021$, raw $p = 0.038$; BH-adjusted $p = 0.037\,8$); plain GIN reaches 0.804 7 ($\Delta = -0.025$, raw $p = 0.018$; BH-adjusted $p = 0.033\,0$); ChemBERTa reaches 0.786 7 ($\Delta = -0.043$, raw $p < 0.000\,1$; BH-adjusted $p = 0.000\,2$). All four comparisons survive Benjamini–Hochberg correction. Under the random split the same verdict holds: ChemBERTa reaches 0.912 1 ($\Delta = -0.031$, raw $p < 0.000\,1$), GIN 0.909 8 ($\Delta = -0.034$, raw $p < 0.000\,1$), GIN–TFP 0.908 4 ($\Delta = -0.035$, raw $p < 0.000\,1$), and GIN–TNE

Table 1: Mean test ROC AUC ± SD of the five per-seed means (each seed averages five fold-level test AUCs). The underlying analysis contains 25 fold–seed replicates per split. The displayed p-values are raw paired tests on the five per-seed means versus ECFP4–RF under scaffold; Benjamini–Hochberg correction across the four comparisons gives adjusted p-values of 0.0330 (GIN), 0.0330 (GIN–TFP), 0.0378 (GIN–TNE), and 0.0002 (ChemBERTa).

Model	Random	Scaffold	Δ vs. ECFP4 (scaffold, raw p)
ECFP4–RF (baseline)	0.9433 ± 0.0003	**0.8300** ± 0.0023	—
GIN	0.9098 ± 0.0022	0.8047 ± 0.0141	−0.025 ($p = 0.018$*)
GIN–TFP (topological fusion)	0.9084 ± 0.0012	0.8138 ± 0.0107	−0.016 ($p = 0.025$*)
GIN–TNE (tensor fusion)	0.8918 ± 0.0018	0.8090 ± 0.0149	−0.021 ($p = 0.038$*)
ChemBERTa	0.9121 ± 0.0012	0.7867 ± 0.0054	−0.043 ($p < 0.0001$***)

0.8918 ($\Delta = -0.052$, raw $p < 0.0001$), each significantly below ECFP4–RF (0.9433); all four comparisons survive Benjamini–Hochberg correction. *No graph-based arm outperforms the fingerprint baseline under either split.*

The result was obtained with a fixed, repeated training protocol (AdamW, validation-based early stopping, best-validation checkpointing, and 25 fold–seed replicates). It is consistent with reports across ADMET/Tox21 endpoints [12] and pretrained-embedding benchmarks [13], in which local substructure encodings captured by ECFP4 and exploited by tree ensembles remain highly data-efficient.

2.3 H2: sequence modelling does not overcome the scaffold gap

ChemBERTa, fine-tuned with per-fold weight reset (see Methods), reaches 0.7867 ± 0.0054 under the scaffold split (SD across five per-seed means; five folds repeated over five seeds; fold-level range 0.724–0.836). Against the 0.8300 fingerprint baseline this is a significant deficit ($\Delta = -0.043$, paired $t(4) = -18.35$, $p < 0.0001$). Under the random split it reaches 0.9121 ± 0.0012, narrowly ahead of GIN (0.9098) but still 0.031 below ECFP4–RF (0.9433). The final scaffold hierarchy is thus:

$$\begin{aligned} &\text{ECFP4–RF } (0.8300) > \text{GIN–TFP } (0.8138) > \text{GIN–TNE } (0.8090) \\ &\qquad > \text{GIN } (0.8047) > \text{ChemBERTa } (0.7867). \end{aligned} \tag{1}$$

The available validation trajectories show how checkpoint selection proceeded over 50 training epochs with an early-stopping patience of 10; these curves describe optimization behaviour rather than establishing the absence of over-fitting (figure 2).

2.4 H3: topological features remain visible in the fusion head despite the ranking deficit

Even when predictive gains are absent, learned fusion weights carry structure that can be examined descriptively. We compute a *salience* score for each descriptor dimension as the mean absolute weight in the fusion projection layer (head_0) over the 25 replicates. For the TFP fusion (78 dimensions: 33 homology-dimension-0, 25 persistent-image, 20 Betti), the *persistent-image block (dims 33–57) has the largest raw mean projection-weight magnitude* (0.0790) versus homology-0 (0.0485) and Betti (0.0547) blocks; the top five dimensions are persistent-image (42, 43, 52, 53, 54). For the TNE fusion (192 dimensions) the top dimensions are 68, 43, 92, 66, 165, and the top-10 % of dimensions account for 17.3 % of total salience (moderate sparsity).

This pattern indicates that topological descriptors are used by the fusion head, but it does not establish feature necessity or causal relevance. On this panel, the descriptor signal does

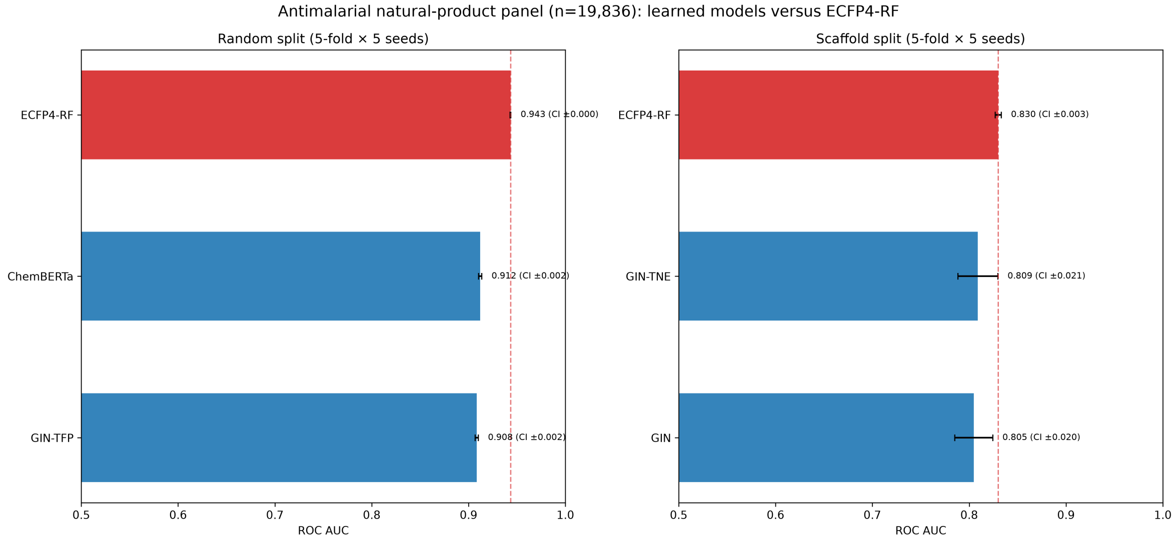

Figure 1: Mean test ROC AUC with 95 % confidence intervals computed from five per-seed means (each mean aggregates five fold-level test AUCs), shown for random (left) and scaffold (right) splits. The underlying analysis contains 25 fold–seed replicates per model and split. Bar annotations report the mean followed by the CI half-width. The ECFP4–RF baseline (red) is not exceeded by any tested GNN or transformer arm.

not translate into a ranking advantage over fingerprints. Thus, projection-weight salience and predictive ranking should be treated as complementary, not interchangeable, summaries.

3 Discussion

The three questions posed at the outset lead to a coherent conclusion. On this panel, learned representations do not exceed ECFP4 under either split; topological fusion narrows the scaffold gap modestly but does not close it; and persistent-image dimensions remain salient in the fusion head without yielding a ranking advantage. Predictive utility and interpretive visibility are therefore separable properties.

3.1 The verdict is scale-independent and panel-specific

The primary result is that the tested pretrained and graph-based models did not improve ROC AUC over ECFP4–RF on this panel, including under scaffold-held-out testing. This conclusion is supported by repeated evaluation, paired comparisons on the five per-seed means, and correction for multiplicity. The transformer result depends critically on fold-independent weight initialization: pretrained weights were reset at the beginning of every fold so that fine-tuning could not transfer between test folds. Future molecular-benchmark studies should therefore state explicitly whether pretrained weights are reset for every fold.

3.2 Why fingerprints hold: representation–task alignment

The consistent ECFP4 advantage is best explained by representation–task alignment rather than model capacity. ECFP4 encodes local substructure environments; for a collection of structurally related natural products, local SAR is the dominant signal, and tree ensembles exploit it efficiently with few thousand labels [21]. Learned graph representations require more data to recover the same local patterns and generalize to scaffold-held-out series only partially [13]. This is consistent with the medicinal-chemistry view that fingerprints plus random forests remain an

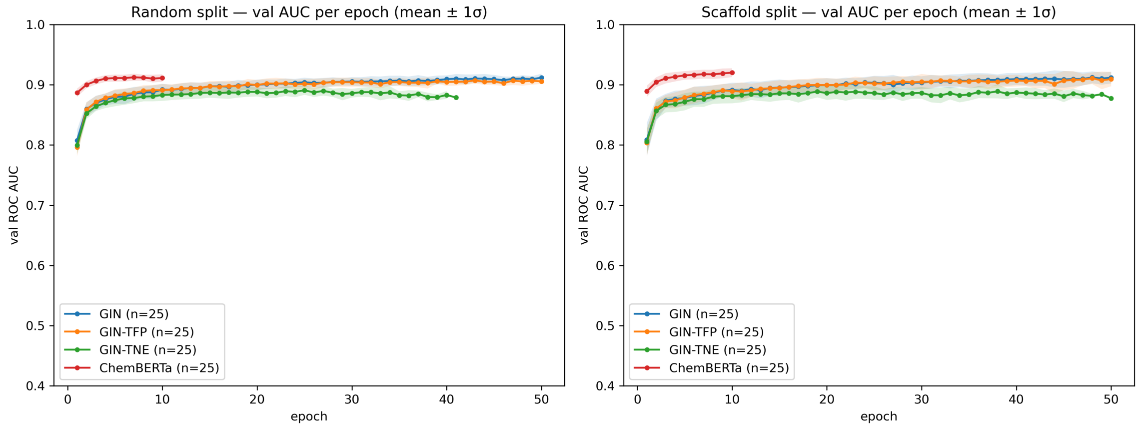

Figure 2: Validation ROC AUC trajectories used for checkpoint selection. Curves are available for GIN under random and scaffold splits, GIN–TFP and GIN–TNE under the scaffold split, and ChemBERTa under the scaffold split; each available arm contains five folds repeated over five seeds. Early stopping with patience 10 selects the checkpoint before test evaluation.

effective early-stage tool [22]. The counterpoint that graph models can be made competitive by training them *on* fingerprints is instructive: ECFP-pretrained GNNs improve out-of-distribution performance on homogeneous ADMET tasks but still underperform on heterogeneous endpoints such as binding affinity [23]—mirroring the fingerprint-centric explanation, since the pre-training transfers the very substructure knowledge fingerprints encode. Our attribution result (H3) adds a nuance: the topological descriptors are *not* uninformative — they carry interpretable geometric signal — but their marginal ranking contribution is small on this panel.

3.3 Relation to prior antimalarial modelling

The molecular collection and fingerprint implementation were previously characterized in a quantum-representation benchmark [20]. That study found no statistically detectable advantage of the simulated quantum kernel over a tuned radial-basis kernel, while the present analysis tests a different question: whether graph or sequence models improve activity ranking on the same chemical collection. Molecular-dynamics studies of selected antimalarial leads have additionally examined resistance-associated targets and mutations [7]; those simulations provide biological context for the screening programme but are not labels or validation data for the present prediction benchmark. The current results therefore stand on their own: topological descriptors are used descriptively by the fusion head, but none of the tested learned representations improves ROC AUC over ECFP4–RF.

3.4 Practical recommendations

For comparable antimalarial NP panels, ECFP4–RF is a strong first-line baseline: it is computationally inexpensive and outperformed every tested neural arm. Graph and sequence models should be evaluated against this baseline under scaffold-held-out testing rather than assumed to improve from model scale alone. Transformer studies should reset pretrained weights for every fold and record the initialization protocol. Topological or tensor-network descriptors can be examined as complementary descriptive features, but their inclusion should not be interpreted as evidence of improved ranking without a paired comparison.

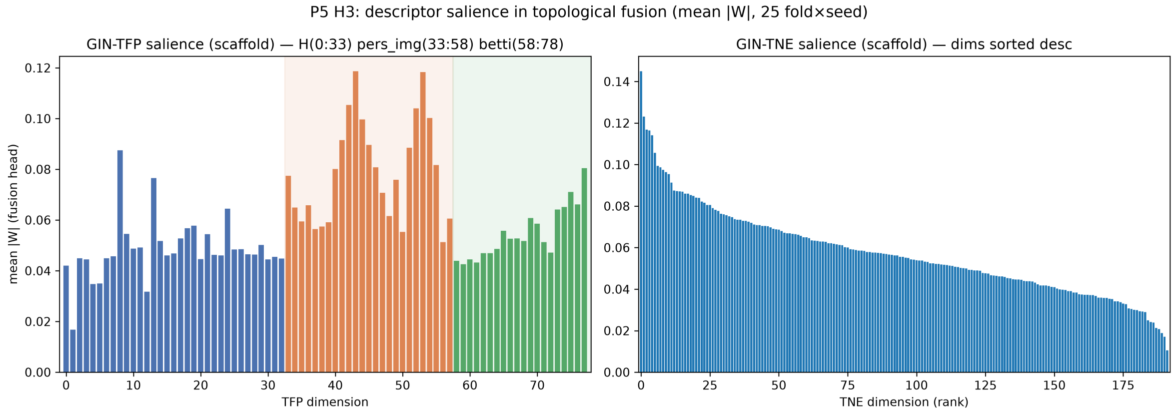

Figure 3: Descriptive projection-weight salience in the topological fusion heads under the scaffold split. Values are mean raw $|W|$ across 25 fold–seed replicates. Left: TFP (78 dimensions), partitioned into homology-0 (dims 0–32), persistent-image (dims 33–57), and Betti (dims 58–77); the persistent-image block has the largest raw weight magnitude. Right: TNE (192 dimensions), shown in descending order. Weight magnitude is interpreted descriptively and is not a causal attribution.

3.5 Limitations

Our panel, while curated and clinically motivated, is a single binary-activity collection assembled from screening data. To test whether our verdict is specific to these computational labels, we repeated the ECFP4–RF versus GIN comparison on a disjoint ChEMBL-derived activity benchmark built from recorded IC_{50}/EC_{50} activities against *Plasmodium falciparum* (target CHEMBL364, 22 267 molecules after excluding 180 compounds overlapping the P5 panel; actives at pChEMBL ≥ 6.0). These are thresholded database records, not newly generated prospective experiments. The fingerprint baseline again wins in this external analysis: ECFP4–RF 0.9547 ± 0.0033 versus GIN 0.9237 ± 0.0040 under the random split ($\Delta = +0.031$, paired $p = 0.0001$) and ECFP4–RF 0.9190 ± 0.0004 versus GIN 0.8843 ± 0.0021 under the scaffold split ($\Delta = +0.035$, paired $p < 0.0001$). Because this external analysis used an independently constructed split rather than the frozen internal fold files, it is a transfer analysis rather than a strict replication of the primary protocol; it nevertheless indicates that the ordering is not unique to the eOS80CH screening labels. The GNN and fusion models use a single-layer prediction head and fixed hyperparameters (hidden dimension 128, 50 epochs, patience 10); alternative architectures or tuning could reduce the observed gap. The external ChEMBL analysis used separate fixed 60/20/20 partitions: a random permutation for the random analysis and a scaffold-group partition for the scaffold analysis, with repeated model initializations rather than the primary frozen five-fold design. Its uncertainty estimates should not be read as interchangeable with the primary five-fold analysis. ChemBERTa was fine-tuned with a fixed learning schedule rather than per-task optimization. The GIN uses 2D molecular graphs without conformational information; 3D-equivariant architectures may perform differently on tasks for which shape is important [19]. The salience measure (mean absolute projection weight) is a first-order attribution that captures correlation between descriptor dimensions and the predictive head, not causation; more sophisticated methods (e.g., integrated gradients, SHAP) would be needed to establish which features are truly necessary for the model's decisions. Our conclusion is that topological features carry interpretable signal in the fusion head, not that they are causally responsible for the predictions. Finally, the scaffold split is a proxy for novelty; prospective experimental validation remains the ultimate test. A recent structural-frontier analysis reinforces this caveat: a scaffold identifier captures only one notion of chemical unfamiliarity, and splits that reserve the sparsest, physicochemically remote scaffold groups inflate error further on ADMET tasks [24]; our verdicts should therefore be read as valid for Bemis–Murcko scaffold held-out series, with harder forms of novelty (e.g., distant scaffold-frontier groups) posing an even steeper generalization test.

4 Conclusions

The central question was whether larger graph and sequence models improve activity prediction beyond ECFP4 when chemical scaffolds are held out. On a frozen panel of 19 836 African antimalarial natural products, ECFP4–RF outperformed the tested GIN, topological-fusion GNNs, and fine-tuned ChemBERTa transformer under both random and Bemis–Murcko scaffold cross-validation. Under the scaffold split, the hierarchy was ECFP4–RF (0.830 0) ¿ GIN–TFP (0.813 8) ¿ GIN–TNE (0.809 0) ¿ GIN (0.804 7) ¿ ChemBERTa (0.786 7); all four learned arms were significantly below the fingerprint baseline after multiplicity correction. The external ChEMBL-derived analysis showed the same ordering for ECFP4–RF versus GIN, while remaining an observational transfer analysis of thresholded activity records. Finally, persistent-image dimensions had the largest projection-weight salience in the fusion head. These findings support using ECFP4–RF as a strong baseline and treating learned topological features as complementary descriptive inputs rather than assuming that model scale improves ranking.

5 List of Abbreviations

ECFP4: extended-connectivity fingerprint, radius 2; **RF:** random forest; **GNN:** graph neural network; **GIN:** graph isomorphism network; **TFP:** topological fingerprint; **TNE:** tensor-network embedding; **ROC AUC:** receiver-operating-characteristic area under the curve; **CV:** cross-validation; **NP:** natural product; **ADMET:** absorption, distribution, metabolism, excretion, toxicity.

6 Methods

6.1 Data and panel

We use a frozen panel of 19 836 molecules with binary antimalarial activity labels from the eOS80CH screening collection, curated and deduplicated by canonical SMILES as described previously [6]. Features are ECFP4 (2 048-bit Morgan radius-2), a 78-dimensional topological fingerprint (TFP; persistent homology, persistent images, and Betti descriptors), and 192-dimensional tensor-network embeddings (TNE). All features are aligned to the same molecule index.

6.2 Splits

Two protocols were evaluated, each with five folds repeated over five seeds (25 fold–seed replicates). (i) *Random*: stratified five-fold cross-validation. (ii) *Scaffold*: molecules were grouped by Bemis–Murcko scaffold before fold assignment; the held-out test scaffolds are disjoint from the training scaffolds. Validation molecules were sampled from the non-test portion and therefore can share scaffolds with training molecules; early stopping consequently assesses optimization on the training chemical distribution, whereas the held-out test fold provides the scaffold-shift estimate. The split indices are released with the study.

6.3 Models

ECFP4–RF: scikit-learn RandomForest on ECFP4, the reference baseline. **GIN**: graph isomorphism network (128 hidden units, mean-pooling readout, single head) trained on molecular graphs (atom/edge features from RDKit). We select GIN as the representative GNN architecture following MolGraphBench [12], which identified GIN as the optimal GNN for molecular regression across multiple benchmarks; GCN, GAT, and GraphSAGE were evaluated in that study and did not consistently outperform GIN, so we omit them to avoid redundancy. **GIN–TFP / GIN–TNE**: the same GIN with the topological descriptor vector concatenated at the head ($\text{head}_0 = \text{Linear}(128 + n_{desc} \to 128)$), enabling the salience attribution. **ChemBERTa**: pre-trained SMILES transformer, fine-tuned with cross-entropy and independent pretrained-weight initialization for every fold. All deep models trained on a single NVIDIA RTX A4000 GPU with AdamW, early stopping on validation AUC (patience 10 of 50 epochs), and best-val checkpoint selection.

6.4 Fold-independent transformer initialization

For each fold, the pretrained ChemBERTa state is restored before fine-tuning, so no fold receives weights updated on another fold. This independence is essential because cross-fold weight transfer can inflate test AUCs by up to 0.15. All transformer results use independent fold initialization and the same training schedule.

6.5 Statistics

Per model and split, we retain the 25 fold–seed AUCs and summarize them as five per-seed means, each averaging the five fold-level test AUCs. Model-versus-baseline comparisons use paired t-tests on these five per-seed means ($df = 4$), rather than treating the 25 fold–seed values as independent observations. Table values are mean ± SD across the five per-seed means; the raw two-sided p-values are corrected with Benjamini–Hochberg separately across the four scaffold comparisons and the four random comparisons. Under scaffold this yields adjusted p-values of 0.0330 (GIN), 0.0330 (GIN–TFP), 0.0378 (GIN–TNE), and 0.0002 (ChemBERTa); under random all four arms remain below the baseline after correction. The significance threshold is $\alpha = 0.05$ and effect sizes are reported as mean differences. Salience is aggregated across the 25 replicates as the per-dimension mean absolute projection weight.

7 Availability of data and materials

All code, frozen splits (random and scaffold, five seeds each), per-fold results, salience data, figure scripts, and the 19 836-molecule panel are available under the MIT licence in the public GitHub repository (`https://github.com/NanaEngo/Malaria_codesV2`). Trained checkpoints and feature tables are included; reproducibility is supported by the frozen split files and the pinned environment.

Secondary scaffold-only replication. An independent scaffold evaluation (five folds × five seeds) produced all 25 fold–seed records and a mean ROC AUC of 0.7908 ± 0.0058 when summarized across five per-seed means; the primary scaffold estimate was 0.7867 ± 0.0054 under the same summary. The primary random-split estimate was 0.9121 ± 0.0012; no independent random-split replication was performed.

8 Author contributions

MVST conceived the study, designed the benchmark protocol, implemented the GNN/transformer pipelines, and wrote the manuscript. All authors contributed to data analysis, reviewed the results, and approved the final version. (CRediT roles and full author affiliations are listed on the title page.)

9 Competing interests

The authors declare no competing interests.

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11 Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

12 Consent for publication

Not applicable. No individual person's data (including images, videos, or identifiable details) appear in this manuscript.

13 Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. Computational results, statistical analyses, and all quantitative claims were generated and verified by the authors; the public repository contains the currently released artefacts.

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